APPENDIX

# Parameter tables

IN THIS CHAPTER

## A.1 Default setpoints

**Table A.1.** System performance defaults (SonixED skid v2025-A7)

| Parameter | Value |
|---|---|
| Flowrate per stack | 0.5–10 $m^3/h$ |
| System flow per skid | up to 30 $m^3/h$ |
| Recovery rate | up to 98% |
| Operating pressure | 0.5–2.5 bar |
| Temperature range | 5–55 ℃ |
| Specific energy | 0.6–2.0 $kWh/m^3$ |
| Skid dimensions (l×b×h) | 3040×1340×2020 mm |
| Incoming line voltage | 3×400 V AC |
| Max. current | 25 A |
| Valve voltage | 24 V DC |
| Control voltage (DC bus) | 24 V DC |
| Dosing SET 1 (min) | 4.0 mA / 0 impulses/min |
| Dosing SET 2 (max) | 20.0 mA / 180 impulses/min |

**Table A.3.** SalinBloc™ container throughput by format (source: HydroVolta brochure)

| Parameter | 20 ft unit | 40 ft unit |
|---|---|---|
| Flow rate | 2–30 $m^3/h$ | 15–60 $m^3/h$ |
| Approx. daily throughput (22 h/day operation) | 44–660 $m^3/day$ | 330–1,320 $m^3/day$ |

Feed range for both formats: brackish groundwater, 1,500–20,000 ppm TDS. TODO: no container weight figure was found in any source reviewed, including the container general-arrangement drawings themselves (their title blocks carry no weight/dimension text field, and any dimension lines are vector geometry, not extractable text — a human must open these drawings visually). Obtain the weight from the container manufacturer or a loaded-unit weighbridge reading before publishing lifting/foundation guidance.

## A.2 Site-specific settings

TODO: no site-specific commissioning/setpoint record was found; use this table to capture the as-commissioned values for a specific installation.

**Table A.5.** Site-specific settings template

| Parameter | Value |
|---|---|
| | |

## A.3 Alarm limits

The P&ID (appendix B) shows instrument spans used as default alarm spans at each skid position: pressure 0–2.5 bar, conductivity 0–200 mS/cm, with high/low alarm tags (PSA, CEA) at every position. HydroVolta's ThingsBoard warning-code table (a real, deployed alarm definition list, source: `08_IT/01_Thingsboard/Warning table`) gives concrete numeric examples for several alarm classes:

**Table A.7.** Representative alarm/warning codes (source: HydroVolta ThingsBoard warning table)

| Code | Class | Condition |
|---|---|---|
| W001 | General | HV-Skid not powered |
| W002 | General | Ventilation off |
| W011–W013 | Chemical | Antiscalant / $H_2SO_4$ / HCl tank empty |
| W101–W110 | Level | Main diluate/concentrate/electrolyte/breaking/client tank level low or high (e.g. diluate below 40 cm or above 90 cm; electrolyte below 20 cm or above 80 cm) |
| W111–W118 | Level | Buffer tank low-low / high level |
| W201–W214 | Electrical | Stack stage 1–7 voltage below or above system specification |
| W300 | Flow | Diluate flow below 500 l/hr |
| W301 | Flow | Concentrate flow below 500 l/hr |

TODO: this table only samples the documented code ranges; the full warning table has ∼57 defined codes and should be transcribed in full by HydroVolta engineering, together with the exact level/pressure setpoints per tank size, once the standard SalinBloc™ tank configuration (chapter 15) is finalized — these codes come from a specific pilot project's tank/stage configuration and the numeric thresholds may not transfer directly. See the alarm tag list in chapters 6, 12 and 15 for additional alarms.